

Jyot Antani, PhD

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Expertise: Optical Microscopy, Single Particle Tracking, Microbial Biophysics,
Phage-Bacteria Interactions, Phage Therapy, Phage Technology

Education

Aug 2016- May 2021	PhD, Pushkar Lele Lab , Department of Chemical Engineering, Texas A&M University (US) Dissertation: Sensory Functions of the Bacterial Flagellar Motor
Jul 2012- May 2016	Bachelor of Technology with Honours in Chemical Engineering Indian Institute of Technology Bombay (IIT Bombay), Mumbai (India)

Advanced Microscopy Training

Optical Microscopy & Imaging in the Biomedical Sciences (OMIBS)

Marine Biological Laboratory, Woods Hole Massachusetts (US)

2023-Now	Research Facilitator (Course Faculty)
Aug 2022	Course Participant

- Basic principles of microscopy from academic and industrial microscope builders and inventors including...

Light path, image formation, and detection	Photophysics of fluorescence, dyes, and fluorescent proteins
Dynamic Imaging Techniques (FRET, FLIM, FCS)	Point Spread Functions, Deconvolution, & Confocal Microscopy
Multiphoton & Light Sheet Microscopy; Adaptive Optics	Super-resolution Techniques: STED, STORM, SIM
- Hands-on practice with over 40 microscopes including the following modalities...

Phase-contrast Microscopy	Differential Interference Contrast (DIC)	Polarized Light Microscopy
Epifluorescence	Forster Resonance Energy Transfer	Fluorescence Lifetime Imaging
Confocal Microscopy	Total Internal Reflection Fluorescence	Deconvolution
Multiphoton Microscopy	Light Sheet Microscopy	Super-resolution: STED, STORM, SIM

Post-PhD Experience

Jun 2021- Now	Mar 2025 - Now	Associate Research Scientist
	Jun 2021 - Mar 2025	Postdoctoral Research Associate
	Yale University (US): Quantitative Biology Institute ; Center for Phage Biology & Therapy Paul Turner Lab , Department of Ecology and Evolutionary Biology In collaboration with Thierry Emonet Lab , Molecular, Cellular, and Developmental Biology	

Academic Positions

Jul 2025-Jun 2028	Editor, <i>mSystems</i> , an American Society for Microbiology journal
May 2025-Now	Executive Editor, Journal of Life Sciences
2024-2026	Early Career Editorial Board Member, <i>The FASEB Journal</i>

Patent

Turner PE, [Antani JD](#), Emonet T. Methods of high-throughput quantification of virus particle attachment.
Patent Application (2025) PCT/US25/49613.

Key Publications

* equal contribution ✉ corresponding authorship

[Full list on Google Scholar \(Link\)](#)

[Antani JD](#)^{*}, Ward T, Emonet T, Turner PE (2024) High-throughput quantification of virus particle attachment to host bacterial cells. *PNAS, USA*. doi: [10.1073/pnas.2410905121](https://doi.org/10.1073/pnas.2410905121)

[Antani JD](#)^{*}, Theroux A, Avgidis F, Scharf B, Emonet T, Turner PE. Flagellotropic bacteriophage selects for evolutionary remodeling of cell motility and chemotaxis in *Escherichia coli*. *bioRxiv*. doi:[10.1101/2025.05.06.652435](https://doi.org/10.1101/2025.05.06.652435) Under review at *Science Advances*.

- Oromí-Bosch A*, Antani JD*, Turner PE (2023) Developing phage therapy that overcomes the evolution of bacterial resistance. *Annu Rev Virology*. doi: [10.1146/annurev-virology-012423-110530](https://doi.org/10.1146/annurev-virology-012423-110530)
- Antani JD[✉], Turner PE. Rage Against the Mean: A Perspective on Measuring Fitness of Individual Phage Particles. In press at *npj Viruses*.
- Antani JD, Gupta R, Lee AH, Rhee KY, Manson MD, Lele PP (2021) Mechanosensitive recruitment of stator units promotes binding of the response regulator CheY-P to the flagellar motor. *Nature Communications*. doi: [10.1038/s41467-021-25774-2](https://doi.org/10.1038/s41467-021-25774-2)
- Antani JD, Sumali AX, Lele TP, Lele PP (2021) Asymmetric random walks reveal that the chemotaxis network modulates flagellar rotational bias in *Helicobacter pylori*. *eLife*. doi: [10.7554/eLife.63936](https://doi.org/10.7554/eLife.63936)
This work helped secure an R01 research grant totaling over \$1.3 million from the National Institute of General Medical Sciences, National Institute of Health. <https://www.eurekalert.org/news-releases/945164>
- Sikkema A, ..., Antani JD et al. (2026) A fully synthetic Golden Gate Assembly system for engineering a *Pseudomonas aeruginosa* phiKMV-like phage. *PNAS, USA*. doi:[10.1073/pnas.2525963123](https://doi.org/10.1073/pnas.2525963123)
A collaborative synthetic biology project with New England Biolabs, a leading biotechnology company.
- Katiyar A*, Antani JD*, McKee BP, Gupta R, Lele PP, Lele TP (2021) A method for direct imaging of x-z cross-sections of fluorescent samples. *Journal of Microscopy*. doi: [10.1111/jmi.12965](https://doi.org/10.1111/jmi.12965)
- Wong GCL, Antani JD, et al. (2021) Roadmap on emerging concepts in the physical biology of bacterial biofilms: from surface sensing to community formation. *Physical Biology*. doi: [10.1088/1478-3975/abdc0e](https://doi.org/10.1088/1478-3975/abdc0e)
- Antani JD, Shaji A, Gupta R, Lele PP (2024) Reassessing the standard chemotaxis framework for understanding biased migration in *Helicobacter pylori*. *Annu Rev Chem Biomol Eng*. doi: [10.1146/annurev-chembioeng-100722-114625](https://doi.org/10.1146/annurev-chembioeng-100722-114625)
- Katiyar A, Zhang J, Antani JD,..., Lele TP (2022) The Nucleus Bypasses Obstacles by Deforming Like a Drop with Surface Tension Mediated by Lamin A/C. *Advanced Science*. doi: [10.1002/advs.202201248](https://doi.org/10.1002/advs.202201248)
- Ford KM, Antani JD, Nagarajan A, Johnson MM, Lele PP (2018) Switching and torque generation in swarming cells of *E. coli*. *Frontiers in Microbiology*, 9, 2197. doi: [10.3389/fmicb.2018.02197](https://doi.org/10.3389/fmicb.2018.02197)

Pre-prints (published, yet to be peer-reviewed)

- Hernandez CA, ..., Antani JD, Turner PE (2026) Prophage activity shapes the thermal ecology and evolution of a marine bacterial host. *bioRxiv*. doi: [10.64898/2026.02.19.706829](https://doi.org/10.64898/2026.02.19.706829).
- Nasserredine A,..., Antani JD,..., King MC, Horsley V (2026) SUN2 mediates epigenetic remodeling to drive mechanotransduction during skin fibrosis. *bioRxiv*. doi: [10.64898/2026.03.19.712957](https://doi.org/10.64898/2026.03.19.712957)
- Müller DM,..., Antani JD,..., Bollyky PL (2024) Bacterial receptors but not anti-phage defence mechanisms determine host range for a pair of *Pseudomonas aeruginosa* Lytic Phages. *bioRxiv*. doi: [10.1101/2024.04.30.591980v2](https://doi.org/10.1101/2024.04.30.591980v2)
- Kanhirun N,..., Antani JD,..., Pride DT (2024) Identification of a large cohort of *Enterobacter jumbo* phages with broad host ranges across pathogenic Gammaproteobacteria. *bioRxiv*. doi: [10.1101/2024.10.29.620934](https://doi.org/10.1101/2024.10.29.620934)

Publications in preparation

- Targ B, Turner PE, Antani JD[✉]. CPFUcounter: An open-source app for counting bacterial colonies/viral plaques.
- Antani JD[✉], Williams A, An A, Turner PE. Multiscale measurements of interactions between jumbo phages and *Pseudomonas aeruginosa* patient isolates. Presented at *IAS-USA Conference on Bacteriophages, Washington DC, Oct 2025*
- Davidson M, ..., Antani JD, et al. Phage Woody selection of *Salmonella* Typhimurium results in cellulose synthesis mutants. Presented at American Society of Microbiology's 2025 annual Microbe meeting in Los Angeles, CA (USA).

Arenas-Castro H, [Antani JD](#), et al. Patterns of post-pollination prezygotic isolation in a group of monkeyflowerers.

Media mentions

- [Times of India](#) – Amdavadi scientist at Yale aims to combat AMR bacteria using viruses
- 2000-word profile feature in *Divya Bhaskar*, a leading Gujarati-language newspaper, highlighting scientific career following an in-depth interview.
- [YaleNews](#) – New tool offers insights into virus-cell interactions
- [Phys.org](#) – Fluorescence microscopy tracks phage attachment to bacteria in real time
- [ASM Microcosm magazine](#) – Friends and Foes: Microbial Interactions and Infection
- [Texas A&M Stories](#) – To Colonize Different Environments, Bacteria Precisely Tune Their Nanomotors
- [Phys.org](#) – A novel approach to determine how carcinogenic bacteria find their targets
- [Texas A&M Stories](#) – A Novel Approach To Determine How Carcinogenic Bacteria Find Their Targets

Invited Talks/Seminars

Jun 2026	American Society for Microbiology Annual Microbe Meeting 2026, Washington DC (US)
Jan 2026	American Society for Microbiology Connecticut Valley Branch Meeting
Nov 2023	Microbial Treasure Hunt: How Bacteria and Viruses Find their Haven . Departmental Seminar, Chemical Engineering , Indian Institute of Technology Bombay (India)
Oct 2021	Force-regulation of CheY-P-binding to the flagellar motor. Howard Berg Lab, Harvard U (US)

Selected Peer Reviews

[Full list on WebOfScience \(Link\)](#)

<i>Microbiology Spectrum</i> (22)	<i>Physical Biology</i> (4)	<i>Applied & Environmental Microbiology</i> (7)
<i>Nature Microbiology</i> (1)	<i>PLoS One</i> (4)	

Honors & Awards

2026	Yale University's nominee for Blavatnik Regional Awards for Young Scientists
2025	New Investigator Scholarship, International Antiviral Society–USA Conference on Bacteriophages: Biology, Dynamics, and Therapeutics
2023	Top Reviewer 2023, <i>Microbiology Spectrum</i> – an American Society for Microbiology (ASM) Journal
2023	Best Poster Presentation Award, Yale Postdoctoral Association Annual Symposium 2023
2022	MBL Alumni ROCS (Regional Outreach and Communication in STEM) Grant
2022	BioImaging North America (BINA) Professional Development Award - Chan Zuckerberg Initiative
2022	Histochemical Society (HCS) Travel Award
2020	CIRTL Associate Certificate, Center for the Integration of Research, Teaching, and Learning
2015	Best Poster Award, AMAT seminar , Mumbai (India)
2015	Undergraduate Research Award URA01, IIT Bombay
2012	All India Rank 1427 in Joint Entrance Examination for engineering, among 479,651 participants
2008	National Talent Search Scholarship, NCERT, India

Key Conference Presentations

Antani JD , Williams A, Turner PE. Jumbo Phages can effectively kill patient isolates despite slow attachment.	
Oct 2025	Conference on Bacteriophages, International Antiviral Society, Washington, DC (US)
Jun 2025	Microbe- Annual meeting of the American Society for Microbiology, Los Angeles (US)
Apr 2025	American Society for Microbiology Joint Branch Meeting, New England & Connecticut Valley Branches (US)
Apr 2025	Emerging Pathogens Initiative Meeting, Howard Hughes Medical Institute (US)

Antani JD, Ward TJ, Emonet T, Turner PE. Quantitative biology of bacterial virus attachment to host surfaces.

Mar 2024 | [American Physical Society March Meeting 2024](#), Minneapolis (US)
Jan 2024 | [Bacteria versus Phage: the Main Event](#), Princeton Center for Theoretical Science Workshop, Princeton University (US)
Jun 2023 | [Microbe- Annual meeting of the American Society for Microbiology](#), Houston (US)
May 2023 | Yale Postdoctoral Association Symposium, Yale University, New Haven (US)
Sep 2022 | Annual Microbiology Retreat, Yale University, New Haven (US)
Jun 2022 | [Microbe- Annual meeting of the American Society for Microbiology](#), Washington, DC (US)
May 2022 | Yale Postdoctoral Association Symposium, Yale University, New Haven (US)

Antani JD, Theroux A, Emonet T, Turner PE. Bacterial Evolution of Resistance To A Flagellotropic Phage.

Sep 2023 | Annual Microbiology Retreat, Yale University, New Haven (US)
Jun 2023 | [Microbe- Annual meeting of the American Society for Microbiology](#), Houston (US)
Jun 2022 | [Microbe- Annual meeting of the American Society for Microbiology](#), Washington, DC (US)

Antani JD, Lele PP. Modulation of chemotactic signaling in bacteria by mechanical forces.

Mar 2019 | ChEGSA Annual Symposium, Texas A&M University, College Station (US)
Oct 2018 | [American Institute of Chemical Engineers \(AIChE\) Annual Meeting](#), Pittsburgh (US)
Dec 2017 | Texas Branch of American Society for Microbiology, 2017 Fall Meeting, College Station (US).

Antani JD, Lele PP. Asymmetric Swimming and Chemotaxis in *Helicobacter pylori*.

Jan 2021 | [Bacterial Locomotion And Signal Transduction \(BLAST\) XVI Meeting](#), Zoom (online)
Feb 2020 | [Biophysical Society \(BPS\) Annual Meeting](#), San Diego (US)
Jan 2019 | [Bacterial Locomotion And Signal Transduction \(BLAST\) XV Meeting](#), New Orleans (US)

Antani JD, Venkatesh KV. Analysis of Glucose Homeostasis in Humans.

Nov 2015 | [Futuristic Approach for Alternatives to Animal Testing](#), CEFIPRA seminar, Mumbai (India)

Technical Skills

Bacteria, Mammalian cell, Virus Culturing	Assay Development	Optical Microscopy: advanced expertise
Perfusion & Thermal Control Experiments	Microfluidics Experiments	Image Analysis Automation
Genome Analysis/Bioinformatics	Prototyping & 3D-printing	Single Particle Tracking
Electron Microscopy (Negative-stain)	qPCR, Genome editing	

Computational Skills

Image Processing & Analysis:	MATLAB, Python, ImageJ – {Highlights: particle-tracking; segmentation}
Genetic Analysis:	Molecular Biology {SnapGene, Benchling}, Bioinformatics
Design & Development:	SolidWorks, AutoCAD, LabVIEW,
Simulation Packages:	ANSYS Fluent, Aspen Plus, OpenFOAM, COMSOL, CHARMM, VMD
Programming:	MATLAB, Python, C, C++, Fortran, HTML

Special mention: **Yale Biotech Club Datathon 2022**

- Performed analysis on single-cell RNAseq data from patients of Colorectal Cancer and Inflammatory Bowel Disease
- Used [scanpy](#) to study the variability in expression of various genes and identify marker genes associated with the two diseases

Current and Doctoral Research

May 2021- Now | Postdoctoral/current Research: **Host-virus Interaction Biophysics in Microbes**
Evolution of target search by bacteriophages on host bacterial surfaces

- Established a protocol for labeling phages (the viruses of bacteria) with a lysine-binding fluorescent dye
- Visualizing labeled phage (bacterial virus) particles attaching to host cells, through epifluorescence and HILO microscopy
- Set up MATLAB-based statistical analysis of single phage trajectories to quantify attachment dynamics
- Employed Bayesian inference to estimate the diffusive states of phage particles from single trajectories
- Developed a quick high-throughput assay to quantify phage adsorption, applicable for emergency phage therapy
- Compared the attachment dynamics of ancestral and coevolved phage-host pairs
- Expanded this approach to mammalian viruses targeting mammalian cells

Evolution of resistance in *E. coli* against a flagellotropic phage

- Evolved mutants resistant against phage χ , a virus that uses rotating bacterial flagella as receptors
- Trained an undergraduate to carry out the experimental evolution protocol, mentored them in writing a successful fellowship application about this project
- Characterized the genotype and phenotype of the evolved mutants to determine the evolutionary consequences

Grant-Writing

- Wrote several fellowship and grant applications to fund the original research projects

Aug 2016-
May 2021

Doctoral Thesis: [Sensory Functions of the Bacterial Flagellar Motor](#)

Molecular mechanisms of host substrate sensing by bacterial flagellar motors

- Determined the effect of mechanical forces on chemotactic output of the bacterial flagellar motor, with phase microscopy, optical tweezers, and TIRF imaging
- Developed a biophysical model for the regulation of allosteric interactions by mechanical force

Motility, chemotaxis, and random spread of *Helicobacter pylori*, the carcinogenic pathogen

- Devised a microscopy-based technique to experimentally determine the chemotactic activity of *H. pylori*
- Studied the biophysical principles governing chemotactic migration by this bacterial species which employs run-reversal motility – different from the traditionally studied *E. coli*

Design and development of experimental and analytical framework

- Designed 3D-printed imaging chambers for requirements in microscopy experiments, e.g., perfusion, imaging side-views
- Wrote MATLAB-based image analysis codes for tracking motile bacteria, mammalian cell-nuclei, and fluorophores

Additional Research Experience

Jul 2015-
Feb 2016

Machine Learning Classification of Glucose Homeostasis in Diabetic and Healthy Individuals

Venkatesh Lab, Chemical Engineering, IIT Bombay

- Formulated a Hill function-based empirical biochemical model of body glucose concentration following a fixed food-intake, obtained least-squares fits to data acquired from 150+ individuals
- Constructed a classification tree based on fit-parameters using machine learning in MATLAB; established a statistical grouping among Diabetic and Healthy individuals

May-Jul
2015

A Novel Protocol for Coating Lipid Bilayers onto Silver Nanocubes

Hung-Jen Wu Lab, Texas A&M University

- Established a tip-sonication protocol to make small unilamellar vesicles of lipids
- Coated the unilamellar vesicles as supported lipid bilayers onto Silver nanocubes
- Employed a Localized Surface Plasmon Resonance-based nanocube sensor to monitor protein binding on lipid bilayers
- Analyzed binding kinetics of GM1 ganglioside with Cholera Toxin Subunit B

May 2014-
Apr 2015

Simulation-Design of a Chemical Vapor Deposition Reactor

Mahesh Tirumkudulu Lab, IIT Bombay

- Simulated fluid flow around a disk, successfully obtained the pressure coefficients observed experimentally in literature.
- Proposed an unconventional design for uniform coating of parylene polymer onto silicon micro-wafers in a micro-scale Chemical Vapor Deposition reactor.
- Modeled Parylene flow using ANSYS Fluent, enhanced the proposed design.

Jun 2013

Process Improvement in Jaggery-manufacturing Technology

Sanjay Mahajani Lab, IIT Bombay

- Studied units for continuous production and recuperation of Jaggery, the traditional non-centrifugal cane sugar
- Analyzed Jaggery samples for acidity, moisture content, water activity, density, sugar content, & structure (XRD, ESEM)

Key
Course
Projects

Determining Diffusion Coefficient of Microparticles using Particle-Tracking Algorithms

Course: Colloidal Applications and Biomolecular Interactions in Life Science (Spring 2018)

- Employed particle-tracking algorithms in MATLAB to track the Brownian diffusion of submicron-sized particles recorded using phase-contrast microscopy
- Determined the diffusion coefficient of particles by calculating the average mean-squared displacement of an ensemble

Proposal: Evolution of Bacterial Flagellum

Course: Evolutionary Bioinformatics (Spring 2017)

- Proposed a study involving the comparison of flagellar genome across multiple bacterial species which has potential to aid in obstruction of pathogenic motility
- Carried out BLAST alignment analysis for nucleotide and protein sequences of flagellin (flagellar filament) gene as preliminary results

Review: How Deep a Cell Can Sense

Course: *Cell Mechanics (Spring 2016)*

- Carried out a literature survey and reported theoretical, in-vitro, and in-silico studies on quantified investigations about the range of mechanical signaling

On-line Estimation of Concentrations in a Bio-Reactor

Course: *State Estimation (Fall 2015)*

- Generated true states from a model based on substrate uptake, phosphorus repression, and effect of seed culture on glyco-peptide antibiotic production
- Implemented the Simple, Extended, and Unscented versions of the Kalman Filter to estimate the concentrations

Teaching & Mentoring Experience

Fall 2019 | A hands-on workshop for image processing and single particle tracking in MATLAB
Pushkar Lele Lab, Texas A&M University

Undergraduate students mentored

Sep 2024-
Now | [Alex Williams](#) & [Allan An](#), Undergraduate Researchers, Yale University
Interactions between jumbo phages and clinical isolates of *Pseudomonas aeruginosa*

Aug 2022-
May 2023 | [Timothy Ward](#), Undergraduate Researcher, Yale University
Attachment of bacteriophages to bacterial cells: the classical adsorption assays
Best Poster Award winner at [Yale Undergraduate Research Symposium](#), March, 2023

Oct 2021-
Oct 2022 | [Austin Theroux](#), STARS II Science & Quantitative Reasoning Education Scholar, Yale University
Evolution of motile bacteria against a bacteriophage that attaches through the flagellum
Presented a poster about this work at *Microbe ASM* meeting at Washington DC in June, 2022

Dec 2019-
Jan 2021 | [Anita Sumali](#), Gathright Scholar, Biomedical Engineering, Texas A&M University
MATLAB-based quantitative analysis of the run-reversal motility of *H. pylori*; co-author on [Antani et al., eLife, 2021](#)
Currently enrolled in EnMed (MEng/MD dual degree program) at Texas Medical Center, Houston (US)

Apr 2017-
Apr 2018 | Aggie_Challenge – Undergraduate Mentoring, Texas A&M University
Graduate-mentor to a multidisciplinary team of eight undergraduate students employing 3D-printing technology to design scaled-up bacterial models, in order to study the hydrodynamic effects of cell morphology on motility

Teaching Assistant at Texas A&M Chemical Engineering

Spring 2019 | Heat Transfer Operations
Spring 2017 | Bioprocess Engineering
Fall 2016 | Bioprocess Engineering

Teaching, Mentoring, & Management Training

Sep 2021-
Dec 2021 | Auburn University's Preparing Future Faculty Workshop
Interactive workshops consisting panel discussions from faculty and university administrators

Sep 2021 | [Fundamentals of Inclusive Teaching](#)
[Poorvu Center: Graduate and Postdoctoral Teaching Development, Yale University](#)
A 4-part workshop on creating a respectful, supportive, equitable, and transparent classroom environment

Jun 2021 | [People Management Skills](#) – Course, Chartered Institute of Personnel and Development, UK

Aug 2018-
Dec 2019 | [Academy for Future Faculty \(AFF\)](#), Center for Teaching Excellence, Texas A&M University
Faculty Mentor: Dr. Micah Green, Texas A&M University
A program designed for training graduate students as future faculty members, featuring weekly seminars and workshops, and anchored by faculty mentorship

Jul-
Aug 2019 | An Introduction to Evidence-Based Undergraduate STEM Teaching
Muti-university Massive Open Online Course from the NSF-founded CIRTLL network
Eight-week online course on [edX.org](#) catering introduction to effective teaching strategies and the research that supports them. Attended local learning community meetings for the course at Texas A&M University.

Oct 2019 | Improving Research Mentoring Relationships – Introduction to Effective Mentorship
Center for the Improvement of Mentored Experiences in Research (CIMER)
1-day workshop at Texas A&M University, conducted by CIMER from University of Wisconsin-Madison

Leadership & Management Experience

Jul 2025- Now	Councilor, American Society for Microbiology Connecticut Valley branch - Organizing and co-ordinating microbiology activities, serving scientists in Connecticut, Rhode Island, parts of New Hampshire, and parts of Massachusetts - Representing the local branch at the national ASM organization
Jul 2024- Now	Communications Committee Member, American Society for Microbiology Member of a team making strategic decisions for the organization ASM with more than 32,000 members worldwide
Jan 2022- Jun 2024	Communications Committee Manager and Coordinator, Yale Postdoctoral Association (YPA) - Led a team of volunteers to manage website, social media, and weekly + monthly newsletters for the YPA - Organized a drive for T-shirts and jackets benefiting 500+ postdoctoral scientists across university
Oct 2018- Aug 2020	Secretary (2018-19) & Chair (2019-20), Chemical Engineering Graduate Student Advisory Council <i>Artie McFerrin Department of Chemical Engineering, Texas A&M University</i> - Designed the council by-laws as a founding member - Set up channels of communication through surveys and anonymous feedback mechanisms, between grad students and administration in the department - Addressed grad student welfare issues including salary, benefits, mental well-being, work-life balance, student-advisor relationships, facilities, career events, and social events
Apr 2014- Mar 2015	Competitions Manager, AZeotropy 2015 <i>IIT Bombay's Intercollegiate Chemical Engineering Symposium</i> - Pioneered a new model-making competition <i>ChemETimer</i> based on time-precision - Designed a presentation-based competition <i>Swaasthya</i> on engineering-applications to healthcare - Designed <i>Cipher</i>, the online cryptic hunt which received a participation of 400+ - Actualized unconventional chemical engineering quizzing for <i>Chem-O-Philia</i>, national level quiz

Science Outreach

May 2025	Glowing Viruses: How Strong is Their Grip? Wrote a summary article explaining the PNAS 2024 work on visualizing virus attachment to bacteria
May 2024	Flipped Science Fair at Yale University Presented a poster about current research topic to middle school students in the New Haven County, Connecticut.
Mar 2023	Tiny World for Little Humans: Teaching Quantitative Microscopy to Underserved Youth - Taught quantitative microscopy through the use of paper microscopes (foldscope.com) to middle school students at Fair Haven School, New Haven, Connecticut - Outreach project funded by the MBL ROCS- Marine Biological Laboratory Regional Outreach and Communication in STEM
Feb 2022- Sep 2022	Associate Writer & Editor, LabLeaks A biweekly mailing list summarizing recent noteworthy scientific articles to the general audience.
Feb 2022	Exploring Science for Middle School Students Presented doctoral research topic to middle school children in the New Haven County, Connecticut
Jul 2020	Coronasurveys: Monitoring the Incidence of COVID-19 via Open Surveys Volunteered services as the primary translator of the survey to the Gujarati language.